

BHANUDEEPAK NAGUMOTHU

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PROFESSIONAL SUMMARY

Data Engineer with 4+ years of combined experience in data engineering, database development, automation, and analytics. Currently building an Azure-based utility data platform using Python, SQL/T-SQL, Azure Databricks, PySpark, ADLS Gen2, Delta Lake, and Azure SQL to integrate billing and operational data across 20+ campus buildings. Experience includes ETL/ELT, relational and dimensional modeling, data pipelines, incremental processing, data quality, reconciliation, REST APIs, and Power BI.

TECHNICAL SKILLS

Programming & Data Processing: Python, SQL, T-SQL, PySpark, Spark SQL, Apache Spark, pandas

Data Engineering: ETL/ELT, Data Pipelines, Data Ingestion & Integration, Batch & Incremental Processing, CDC, MERGE/Upsert, SCD Type 1/2, Data Transformation, REST APIs

Cloud & Lakehouse: Microsoft Azure, Azure Databricks, ADLS Gen2, Delta Lake, Medallion Architecture, Azure SQL Database, Microsoft SQL Server, Unity Catalog

Data Modeling & Quality: Relational & Dimensional Modeling, Data Warehousing, Star Schema, Fact & Dimension Tables, Schema Design, Data Validation, Reconciliation, Deduplication, Schema Evolution

Analytics & Engineering Tools: Power BI, SQL Server Management Studio (SSMS), Git, GitHub, VS Code, Microsoft Power Automate

EDUCATION

Cleveland State University

Master of Science in Information Systems | GPA: 3.77

Cleveland, OH

Expected Dec 2026

Vaagdevi College of Engineering

Bachelor of Technology in Electronics & Communication Engineering | CGPA: 3.6

India

Aug 2019 – Aug 2023

PROFESSIONAL EXPERIENCE

Data Analyst – Utilities Department

Cleveland State University

Aug 2025 – Present

Cleveland, OH

- Manage electric, steam, chilled water, natural gas, sewer, and water billing, meter, usage, cost, account, and operational data across **20+ campus buildings** and **170+ monthly bill-related items**, supporting reconciliation, reporting, and operational decision-making.
- Design and develop a centralized **Azure SQL Database and Microsoft SQL Server** data platform for buildings, vendors, accounts, meters, bills, usage, charges, and payments using relational modeling, schema design, validation rules, and source-to-target mappings.
- Build **Python and SQL ETL workflows** to ingest, transform, validate, reconcile, and prepare data from vendor bills, spreadsheets, APIs, and operational sources for structured storage and analytics.
- Implement data-quality controls to identify duplicate records, missing billing periods, meter-date gaps, usage discrepancies, unit mismatches, and reconciliation exceptions through automated validation and root-cause analysis.
- Configure secure Azure SQL connectivity, access controls, firewall rules, and workflow automation while supporting downstream **Power BI reporting** and operational analytics.

Cyber Data Review Analyst (Contract)

Vipany Global Ltd

Oct 2023 – Dec 2024

Hyderabad, India

- Reviewed, validated, cleaned, and reconciled **1,000+ records daily**, maintaining high data accuracy through structured quality-control and exception-review processes.
- Improved reliability across **80K+ records** by identifying data inconsistencies, performing root-cause analysis, standardizing records, and documenting recurring data-quality issues.
- Applied structured review and validation procedures to detect missing, inconsistent, and duplicate information while supporting accurate downstream data usage and reporting.

Business Data Analyst

Laxmi Precast Allied Products

Jul 2020 – Dec 2022

Hyderabad, India

- Analyzed financial, sales, customer-order, payment, and operational data to support budgeting, cost visibility, business-performance tracking, and management decision-making.
- Maintained and reconciled structured datasets for sales, customer orders, payments, expenses, and reporting while identifying and correcting data-quality inconsistencies.
- Prepared recurring business and operational analyses to improve visibility into customer activity, payments, costs, and overall business performance.

DATA ENGINEERING PROJECTS

CSU Utilities Azure Data Engineering & Lakehouse Platform

Azure Databricks | PySpark | Spark SQL | ADLS Gen2 | Delta Lake | Azure SQL | T-SQL | Unity Catalog | Power BI

- Architecting an end-to-end Azure data platform to centralize billing, meter, usage, cost, payment, and operational data across 20+ campus buildings and 170+ monthly bill-related items, replacing fragmented spreadsheet-driven workflows.
- Implementing Azure Data Lake Storage Gen2 (ADLS Gen2) and a Medallion Architecture (Bronze/Silver/Gold) to separate raw source data, validated engineering datasets, and business-ready consumption layers.
- Developing transformation pipelines with Azure Databricks, PySpark, Spark SQL, and Delta Lake, including schema validation, standardization, deduplication, source-to-target mapping, and reconciliation.
- Designing incremental data processing using business keys, MERGE/upsert patterns, and SCD Type 1/Type 2 techniques to manage current and historical building, vendor, account, and meter attributes.
- Building reusable data-quality controls for duplicate detection, billing-period gaps, meter-date inconsistencies, unit mismatches, null validation, and source-to-target reconciliation, with rejected records routed for review instead of silently discarded.
- Designing Gold-layer dimensional models and governed datasets for Power BI and operational reporting, incorporating idempotent processing, audit logging, schema evolution, lineage, and role-based access.

CSU ENERGY STAR Automation | *Python, pandas, REST APIs, ETL, Data Validation*

- Built a Python ETL pipeline to ingest, transform, validate, standardize, and prepare recurring multi-utility data for ENERGY STAR Portfolio Manager reporting.
- Processed and validated **6,854+ utility records** through automated test workflows, verifying billing periods, meter dates, usage, costs, units, duplicate records, and data consistency prior to REST API transmission.
- Reduced recurring ENERGY STAR data preparation from approximately **3 workdays to 5 minutes** by automating extraction, transformation, validation, reconciliation, and API-ready data preparation.

Steam Bill Automation System | *Python, pdfplumber, openpyxl, ETL*

- Engineered a Python ingestion and transformation workflow using **pdfplumber** and **openpyxl** to parse semi-structured utility PDFs and convert meter readings, usage, multipliers, billing periods, and charges into standardized Excel datasets.
- Reduced steam-bill data extraction from approximately **1 hour to under 2 minutes (97%)** while preserving existing formulas and established business rules.
- Automated validation and reconciliation for missing meters, zero-usage records, multiplier changes, usage mismatches, and bill-to-workbook discrepancies.

CERTIFICATIONS & TRAINING

- **Microsoft Applied Skills: Migrate SQL Server Workloads to Azure SQL Database**
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
- Oracle Cloud Infrastructure 2025 Certified Foundations Associate
- AWS Academy Cloud Foundations
- AWS Academy Machine Learning Foundations

HONORS & AWARDS

- Monte Ahuja College of Business Merit Scholarship – **\$6,000** | Graduate and Professional Student Association Scholarship – **\$2,000**

LEADERSHIP

Vice President – Graduate and Professional Student Association (GPSA)

Cleveland State University

Aug 2025 – Present

Cleveland, OH

- Lead graduate-student initiatives and collaborate with university leadership and cross-functional stakeholders on programming, planning, communications, and organizational initiatives.